

HOIN JUNG

Purdue University, West Lafayette, IN, USA

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EDUCATION

Purdue University <i>Ph.D. in Electrical and Computer Engineering</i> · Dissertation: Efficient Surgical Interventions for Trustworthy Multimodal Machine Learning Systems	West Lafayette, IN, USA Jan. 2023 – Expected May. 2027
Seoul National University <i>M.S. in Computational Science and Technology</i> · Thesis: Local-Ensemble Graph Collaborative Filtering with Spectral Co-Clustering	Seoul, Korea Sep. 2020 – Aug. 2022
Korea Aerospace University <i>B.E. in Aerospace & Mechanical Engineering</i> · Major of Aircraft System Engineering	Goyang, Korea Mar. 2010 – Feb. 2014

RESEARCH INTERESTS

Trustworthy & Efficient Multimodal Systems

Research Context

- Primary Domain: Trustworthy AI, Multimodal Systems, Retrieval-Augmented Generation (RAG)
- Key Problems: Factual Grounding, Fairness & Debiasing, Uncertainty Quantification (UQ), Hallucination

Research Methodology

- Core Philosophy: Efficient Post-Hoc Methods without Costly Retraining
- Key Techniques: Causal Inference, Latent Feature & Logit Adjustment, Multimodal Explainability

AWARDS AND SCHOLARSHIPS

Bilsland Dissertation Fellowship <i>Purdue University</i> • Awarded to outstanding Ph.D. candidates to support final dissertation completion.	Spring 2027
KDD 2026 Research Track Best Reviewer (370 selected)	Aug. 2026
CVPR 2026 Outstanding Reviewers (Top 5%)	Jun. 2026
Advancing Machine Learning for Scientific Discovery Fellowship <i>Los Alamos National Laboratory</i> • Awarded to select graduate researchers to develop trustworthy and transparent machine learning methodologies for critical scientific applications.	Summer 2026
Purdue University Nominee for Google PhD Fellowship (4 of 32 selected)	Apr. 2026
CVPR 2025 Outstanding Reviewers (Top 5%)	Jun. 2025
Purdue Graduate Student Government Travel Grant	Nov. 2024
NeurIPS 2024 Scholar Award	Oct. 2024
NeurIPS 2024 Spotlight Paper (Top 2.5%)	Oct. 2024
Future Industry Talent Graduate Scholarship <i>Hyundai Motor Chung Mong-Koo Foundation</i>	Fall 2021–Spring 2022
National S&T (Science & Technology) Scholarship <i>Korea Student Aid Foundation</i>	Fall 2010

PUBLICATIONS

1. **H.Jung**, and X.Wang, “Uncertainty-Aware Hybrid Retrieval for Long-Document RAG”, *Empirical Methods in Natural Language Processing (EMNLP)*, Main Conference, 2026. (Accepted)
2. **H.Jung**, and X.Wang, “Beyond Chunking: Efficient Global Pooling for Holistic Long-Document Representation”, *Empirical Methods in Natural Language Processing (EMNLP)*, Main Conference, 2026. (Accepted)
3. **H.Jung**, S.Lu, D.Wang, and X.Wang, “Assessing the Reliability of Image Quality Metrics and Mitigating Quality Bias in Generative Models”, *Conference on Computer Vision and Pattern Recognition (CVPR)*, Findings, pp. 7956–7967, 2026.
4. **H.Jung**, V.Nascimento, H.Liu, X.Wang, C.K.Koh, and D.Jiao, “Generative Antenna Design with Accuracy, Diversity, and Explainability via Dynamic Pseudo-Labeling”, *IEEE Journal on Multiscale and Multiphysics Computational Techniques (JMMCT)*, vol. 11, pp. 246–257, 2026.
5. **H.Jung**, J.Chai, and X.Wang, “Adaptive Logit Adjustment for Debiasing Multimodal Language Models”, *International Conference on Learning Representations (ICLR)*, 2026. (27 pages incl. appendix)
6. **H.Jung**, J.Liu, A.Rao, H.Kim, X.Zhao, A.Chandra, and M.Sarkis, “TVAgent: A lightweight Vision-Language-Model for TV GUI Agent”, *AAAI Workshop on Deployable AI (DAI)*, 2026.
7. T.Jang, **H.Jung**, and X.Wang, “Target Bias Is All You Need: Zero-Shot Debiasing of Vision-Language Models with Bias Corpus”, *International Conference on Computer Vision (ICCV)*, pp. 1935–1946, 2025.
8. **H.Jung**, J.Chai, and X.Wang, “Adversarial Latent Feature Augmentation for Fairness”, *International Conference on Learning Representations (ICLR)*, 2025. (36 pages incl. appendix)
9. H.Lee, **H.Jung**, and S.Bae, “Framing Korea: the role of international student YouTubers in shaping destination perceptions”, *Current Issues in Tourism*, pp. 1–16, 2025.
10. **H.Jung** and X.Wang, “Towards On-the-Fly Novel Category Discovery in Dynamic Long-Tailed Distributions”, *Winter Conference on Applications of Computer Vision (WACV)*, pp. 6795–6804, 2025.
11. **H.Jung** and X.Wang, “Fairness-Aware Online Positive-Unlabeled Learning”, *Empirical Methods in Natural Language Processing (EMNLP)*, Industry Track, pp. 170–185, 2024.
12. **H.Jung**, T.Jang, and X.Wang, “A Unified Debiasing Approach for Vision-Language Models across Modalities and Tasks”, *Neural Information Processing Systems (NeurIPS)*, pp. 21034–21058, 2024. (**Spotlight**)
13. **H.Jung**, V.Nascimento, H.Liu, X.Wang, C.K.Koh, and D.Jiao, “Explainable Planar Multiband Antenna Designer with Wasserstein Generative Adversarial Network”, *IEEE International Symposium on Antennas and Propagation (AP-S)*, 2024. (**Oral Presentation**)
14. **H.Jung**, H.S.Choi, and M.Kang, “Boundary Enhancement Semantic Segmentation for Building Extraction From Remote Sensed Image”, *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, vol. 60, pp. 1–12, 2022.

PAPERS UNDER REVIEW AND MANUSCRIPTS

1. **H.Jung**, and X.Wang, “The Cost of Context: Mitigating Textual Bias in Multimodal Retrieval-Augmented Generation”, *Neural Information Processing Systems (NeurIPS)*, 2026.
2. **H.Jung**, D.DeSantis, A.Murph, J. Strait, and X.Wang, “CausalDrift: A Large-Scale Benchmark for Non-Stationary Causal Discovery Under Temporal Drift”, *Transactions on Machine Learning Research (TMLR)*.
3. **H.Jung**, H.Liu, K.Chao, X.Wang, C.K.Koh, and D.Jiao, “Calibrated Uncertainty-Aware Graph Surrogates for Broadband S-Parameter Estimation of Chiplet Interconnects”, *IEEE Transactions on Components, Packaging and Manufacturing Technology (T-CPMT)*.
4. S.Lu, **H.Jung**, Z.Fang, and X.Wang, “Open-World Diversity Guidance for Fair Image Generation,” *Winter Confer-*

ence on Applications of Computer Vision (WACV), 2027.

5. C.Han, Y.Sim, **H.Jung**, J.Lee, H.Lee, YS.Kang, S.Woo, G.Kim, HW.Park, and M.Jun, “IMPACT: Industrial Machine Perception via Acoustic Cognitive Transformer”, *Manuscript*, 2026.
6. X.Liu, R.Zhang, W.Yu, S.Xiong, H.Liu, F.Wu, **H.Jung**, M.Fredrikson, X.Wang, and J.Gao, “The Vision Wormhole: Latent-space Communication in Heterogeneous Multi-Agent System”, *Manuscript*, 2026.

TEACHING EXPERIENCE

ECE 695 Machine Learning in Bioinformatics and Healthcare

Guest Lecture

Purdue University, West Lafayette, IN

Fall 2025

- Designed and delivered a lecture on “Post-Training and Inference-Time Approaches for ML in Healthcare,” introducing advanced efficiency techniques to graduate students.

ECE 570 Artificial Intelligence

Teaching Assistant, Electrical & Computer Engineering

Purdue University, West Lafayette, IN

Fall 2024, Spring 2025, Spring 2026

- Held office hours and led student projects across three course sections, serving a total of 804 students

Computer Literacy & Programming (Python)

Instructor, Language Education Institute

Seoul National University, Seoul, Korea

Mar. 2021 – Jul. 2022

- Designed and delivered a Python programming course for beginner-level students for three semesters
- Earned an average course evaluation score of **4.91/5.00** across 14 sections

L0444: Basic Computing (Python)

Teaching Assistant, Faculty of Liberal Education

Seoul National University, Seoul, Korea

Spring 2021, Spring 2022

- Led weekly lab sessions for 50+ students each semester

WORK EXPERIENCE

Los Alamos National Lab (LANL)

Advancing Machine Learning for Scientific Discovery Fellowship (Graduate Internship)

Los Alamos, NM, USA

May 2026 - July 2026

- Developed a computational benchmark for evaluating causal-discovery methods on nonlinear, nonstationary time-series data with changing causal structures.
- Implemented and evaluated machine-learning and statistical causal-discovery baselines in Python using standardized windowing protocols and graph-recovery metrics.
- Designed experiments to analyze the effects of causal drift, observation-window length, and model assumptions on causal-graph identification performance.

Heterogeneous Integration Design Institute

Research Assistant, Elmore ECE Emerging Frontiers Center

West Lafayette, IN, USA

Jan. 2023 – Present

- Designed an automatic generative designer for multi-band planar antenna
- Engineered an explainable model for the ML-based EM simulation via SHAP values

Samsung Research America

Research Scientist Intern

Irvine, CA, USA

May. 2025 – Aug. 2025

- Conducted research on AI-driven Smart TV solutions in collaboration with senior engineers
- Developed an automatic keyboard navigation system using vision-language models (VLMs), specifically optimized for commercial deployment

Samsung Electronics

Engineer, R&D Team, Department of Digital Appliance

Suwon, Korea

Aug. 2017 – Aug. 2020

- Developed the thermo-fluid performance of freezing system for brand-new refrigerator
- Analyzed and optimized refrigeration cycle control system to reduce the power usage

ROK Air Force

Lieutenant, Aircraft Maintenance Officer, The 19th Fighter Wings

- Managed aircraft line maintenance and administered ground safety department for the military base

Chungju, Korea

Jun. 2014 – May. 2017

ACADEMIC SERVICE

Conference Reviewing & Program Committees

- Area Chair: IEEE ICASSP (2026)
- Program Committee: AAAI (2024, 2025, 2026), IEEE BigData (2025)
- Reviewer: ICLR (2026), CVPR (2025, 2026), WACV (2026), NeurIPS (2025, 2026), SafeMM-AI Workshop (ICCV 2025), ECCV (2024), KDD (2024, 2026)

Journal Reviewing

- IEEE Journal on Multiscale and Multiphysics Computational Techniques (JMMCT)
- Transactions on Machine Learning Research (TMLR)
- IEEE Transactions on Geoscience and Remote Sensing (TGRS)

University Service

- Grant Review and Allocation Committee, Purdue Graduate Student Government

LEADERSHIP

Mentor, Purdue ECE G-LaMP (Graduate Leadership and Mentorship Program) (2025-2026)

Co-Chair, ICON Student Research Conference, Purdue University (2026)

Vice President, Students Government, Korea Aerospace University (2013)

PRESENTATIONS

1. “Adaptive Logit Adjustment for Debiasing Multimodal Language Models” Mar. 2025
Poster, Purdue ECE Open House Symposium
2. “A Unified Debiasing Approach for Vision-Language Models across Modalities and Tasks” Dec. 2024
Spotlight Poster, Neural Information Processing Systems (NeurIPS 2024)
3. “An Efficient and Unified Debiasing Approach for Vision-Language Models across Modalities and Tasks” Jul. 2024
Lightning Talk, Fast Machine Learning for Science Conference 2024
4. “Explainable Planar Multiband Antenna Designer with Wasserstein Generative Adversarial Network” Jul. 2024
Oral, 2024 IEEE International Symposium on Antennas and Propagation
5. “Boundary Improvement Module for Binary Semantic Segmentation in Remote Sensing” Jun. 2021
Oral, Korean Society for Industrial and Applied Mathematics (KSIAM)
6. “Segmentation model for tracking building in satellite imagery” Nov. 2020
Poster, Korean Society for Industrial and Applied Mathematics (KSIAM)